

The Impact of Data Science on the Financial Industry and Business Finance

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Introduction

In the 21st Century, Data Science (DS) has become more and more prevalent in every sector of society, from education to finance. With cutting advances in technology and software, DS has been indispensable for making critical decisions for businesses. One such field in DS and Artificial Intelligence (AI), Machine Learning (ML) and Deep Learning (DL) has become more and more important in predicting and analysing problems or events. As such, learning and knowing about these algorithms are essential for businesses to remain successful in the current age.

As such, This assignment will explore the differences between ML and DL, how financial institutions leverage DS to succeed and how students can use DS to learn about Business concepts such as capital budgeting techniques.

Machine Learning And Deep Learning

ML, as a branch of AI, allows computers to imitate the way that humans learn, to perform tasks autonomously [1], allowing them to use pre-existing data to learn about characteristics in the data infer results automatically from unknown data in the future. This includes a wide number of different algorithms and techniques, such as regression and clustering to create mathematical models that can classify and predict new data.

In comparison, DL, is a subset of ML, using neural networks (NN) [2] to learn about the features of the input data. These NN mimic how the human brain works, using layers of interconnected neurons to train algorithms. Some examples of how neural networks can be used are in Computer Vision, Natural Language Processing and Translation.

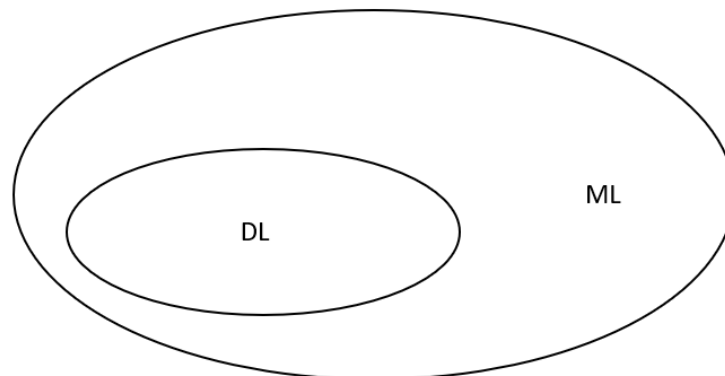


Fig 1 Diagram of the difference of DL and ML using set theory

The main differences of ML and DL is how they are created. ML are manually engineered by data analysts to extract the features from a raw data. However, DL does it without human supervision [3]. In addition, features that may be hard for ML to detect, such as those with complex tasks, would be better for DL. However, deep learning also requires much more data than machine learning [4], thus the computational and the financial cost would often be more expensive than ML.

One example of the difference in how ML differs from DL is how it is used in stock price predictions. Traditional ML algorithms, such as Random Forest or Support Vector Machines (SVMs) operate on structured data to extract features such as simple moving average (SMA), convergence divergence moving average (MACD), and relative strength index (RSI) [5]. However, this approach introduces two critical limitations. First, the model's performance becomes highly dependent on the quality and quantity of data [6]. Which may fail to capture complex, non-linear patterns inherent in financial markets. Second, such models require large volumes of labeled historical data for training, and their accuracy diminishes significantly when applied to volatile environments.

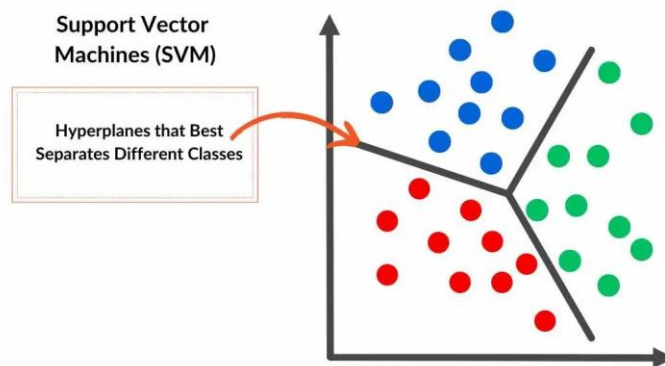


Figure 2 Diagram of SVM [7]

In contrast, DL networks, such as Long Short-Term Memory (LSTM) Networks, process raw, sequential data without to capture the dependencies and dynamics of data, such as using intraday price movements, news sentiment, or earnings call transcripts to capture nonlinear dependencies and temporal dynamics. Chau [8] writes that social media engagement data can also be used for sentiment analysis, making them particularly effective in high-frequency trading or sentiment-driven market prediction. Nevertheless, DL's efficacy is contingent upon large-scale datasets and substantial computational resources. Another disadvantage is how its

heuristic nature makes it unexplainable causes it to be challenging to use in the finance sector.

ML and DLs distinct yet interconnected approaches to artificial intelligence, each with unique strengths and limitations. ML relies on manual feature engineering and structured data, employing algorithms like Random Forest or SVMs to tackle tasks such as stock price prediction using predefined indicators. While effective for simpler patterns, its performance hinges on data quality and struggles with non-linear, dynamic environments.

DL, a subset of ML, leverages neural networks to autonomously extract features from raw, unstructured data (e.g., intraday prices, social media sentiment), excelling in capturing complex dependencies through architectures like LSTMs. However, this capability demands vast datasets, significant computational resources, and sacrifices interpretability—a critical drawback in sectors like finance where explainability is essential.

The choice between ML and DL ultimately depends on the problem's complexity, data availability, and the need for transparency. ML remains practical for tasks with structured data and limited resources, while DL thrives in handling intricate, high-dimensional problems at the cost of scalability and explainability. Balancing these trade-offs is key to deploying effective AI solutions across diverse applications, allowing firms to maximize their benefits.

Data Science in Financial Institutions

Data Science (DS), a field related to AI, ML and Statistics, has become more and more prevalent in financial institutions to gain an advantage. With the internet and computers becoming more and more involved around the world, financial institutions can collect substantial data, managing and governing it to assist companies to make decisions, such as in risk management, stock trading, fraud detection and many other fields.

DS is used by financial services firms such as JPMorgan for performing fraud detection in payments. Fraud in financial sector has increased as technology advances, so data science has become extremely critical to prevent financial fraud. Utilising AI models, JPMorgan perform payment validation screening for the past few years, reducing fraud and improving customer experience, reducing account validation rates by 15-20% [9]. In addition, Machine Learning and Data Analytic techniques are utilised to analyse transaction data to identify unusual patterns and improve fraud detection strategies [10].

Another use case of DS is in Risk Management and Decision Making. Firms such as Deloitte provide risk data management. Deloitte has increased the use of advanced analytics for internal auditing from 7 to 21% in 2018 [11], incorporating it into management and administration. Predictive decision making also uses data analysis to warn firms of potential problems and risks, allowing firms to see indicators that may suggest firms need to take action to prevent failure [12].

Sentiment analysis studies focus on analyzing people's sentiments, attitudes, emotions and opinions about events and facts [13]. This emerging technology is utilised by financial institutions to make decisions. Through analysis of different data, such as customer data or public information such as news and social media, firms can find out public sentiment or information about different events, which can be used for making important decisions and predict events. Du et al. [14] notes that "One of the most attractive applications is to perform forecasting in financial markets", allowing trading firms and hedge funds to make more informed trades. Another example is to analyze customer feedback or data, allowing firms to make better decisions when maximizing profit.

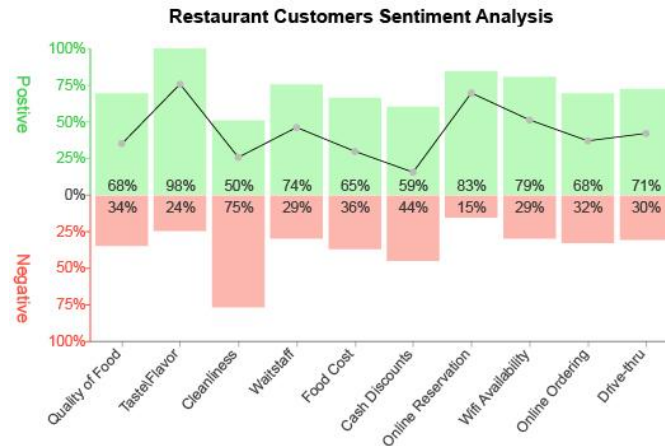


Figure 3 Example graph of customer analysis for a restaurant [15]

Data Science has undeniably transformed the financial sector, enabling institutions to harness the power of AI, ML, and advanced analytics to address critical challenges and drive innovation. By leveraging vast datasets, firms like JPMorgan and Deloitte have demonstrated the tangible impact of DS in fraud detection, risk management, and predictive decision-making—reducing fraud rates, optimizing audits, and preempting risks. Sentiment analysis further extends these capabilities, allowing institutions to decode market trends and public sentiment from diverse data sources, enhancing strategic trading and forecasting. As DS technology evolves, the integration of data-driven approaches will remain pivotal for financial institutions seeking agility, operational efficiency, and a competitive edge in an increasingly complex and data-rich world. The adoption of DS not only safeguards financial systems but also ensures firms can do smarter, faster, and more customer-centric decision-making. As such, learning about DS and how it can be applied in the Finance and Business Sector is critical for success.

Learning Capital Budgeting Techniques Using Data Science

The advancement of data science (DS) has significantly reduced barriers to mastering capital budgeting techniques, enabling students to evaluate investments, expenditures, and long-term projects. By integrating AI and DS, learners can interactively explore methodologies such as Net Present Value (NPV) analysis and Real Options Analysis (ROA). Below, we illustrate three examples of how DS enhances pedagogical outcomes through personalized learning, computational modeling, and dynamic visualization.

Large Language Models (LLMs) such as GPT-4 or Llama to learn about Capital Budgeting by leveraging Generative AI (GenAI) for personalized education. These models can generate tailored explanations and generate personalized feedback and addressing student misconceptions [16]. For instance, a student struggling with the concept of discount rates could prompt an LLM to explain its role in NPV calculations. Additionally, GenAI can simulate financial scenarios, offer interactive quizzes, and adapt content to different skill levels, enhancing comprehension. Using GenAI, students can have an engaging and unique way of learning Capital Budgeting, ensuring that their understanding of these financial budgeting techniques are complete.

Programming languages like Python and R enable students to construct robust, data-driven capital budgeting models using real-world financial data. Students can use public datasets to analyze different projects using programs to calculate NPV. For ROA, students can use QuantLib, a *free/open-source* library for modeling, trading, and risk management to create simulations [17]. For example, students can code Monte Carlo Simulations, a type of computational algorithm that uses repeated random sampling to obtain the likelihood of a range of results of occurring [18] to quantify the value of the “option to abandon” or other indicators. Using DS, students can use real world datasets and apply their knowledge in real-life situations, allows them to make data-backed decisions and gain experience in doing Capital Budgeting.

The visualisation of financial data can be used by students to understand how different features of data that would affect Capital Budgeting outcomes. By using graphs and tables, data visualization tools provide an accessible way to see and understand trends, outliers, and patterns in data [19]. Tools such as Tableau and Plotly allow students to plot NPV

graphs. When modifying variables such as discount rate or cash growth, students can see how the change in these variables would change the NPV graph. Data visualization could also be used by students in presentations to share their findings to their fellow classmates and students, highlighting their understanding and enhancing the learning of students.



Table 4 Graph showing NPV, FV and Sales in Tableau [20]

The use of DS in educating students about Capital Budgeting Techniques allow students to learn both theoretical and practical concepts through GenAI, Programming Languages and Data Visualisation tools. Engaging and learning about these tools would equip students with practical, real-world knowledge about Capital Budgeting and use them to face Financial Challenges from an analytical perspective, enhancing their knowledge and understanding.

Conclusion

In conclusion, DS has become more and more important in the 21st century, particularly in Finance and Education. DS separates ML and DL through their methodologies: ML employs structured data and manual feature engineering for transparent tasks like fraud detection, while DL leverages neural networks to autonomously analyze unstructured data, excelling in more complex domains. Financial institutions strategically integrate these technologies, balancing DL's predictive power with ML's interpretability to address regulatory and operational challenges. Concurrently, DS reshapes business education by transforming capital budgeting into an interactive, data-driven discipline. Students can utilize GenAI for personalized learning, program financial models in Python/R with real-world datasets, and visualize outcomes via tools like Tableau, bridging theoretical concepts with practical applications. As DS advances, innovations in GenAI and DL will deepen their integration across sectors. Organizations adopting DS will sustain competitiveness through enhanced decision-making, while equipping future professionals to tackle evolving challenges with analytical rigor. Thus, DS emerges as a foundational pillar of innovation and strategy in an increasingly data-centric, interconnected

world, poised to drastically change both finance and education in the near future.

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